



Matrix

$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \end{bmatrix}$ 1 row 3 column
 $B = \begin{bmatrix} 3 \\ 4 \\ -5 \end{bmatrix}$ 1 column 3 row

$AB = [2(3) + 5(4) + (6)(-5)]$
 $= [6 + 20 + (-30)]$
 $= [-4]$ 1 row 1 column

$BA = \begin{bmatrix} 6 & 15 & 18 \\ 8 & 20 & 24 \\ -10 & -25 & -30 \end{bmatrix}$ 3 rows 3 column

$3 \times 5 = 15$
 $5 \times 3 = 15$

$1 \times 1 \quad AB = A \times B$
 $3 \times 3 \quad BA = B \cdot A$

$A = \begin{bmatrix} 1 & 4 & -2 \\ 3 & 5 & -6 \end{bmatrix}$ 2 row 3 column
 $B = \begin{bmatrix} 5 & 2 & 8 & -1 \\ 3 & 6 & 4 & 5 \\ -2 & 9 & 7 & -3 \end{bmatrix}$ 3 row 4 column

$AB = \begin{matrix} A & B \\ 2 \times 3 & 3 \times 4 \\ \hline 2 \times 4 \end{matrix}$ same

$BA = \begin{matrix} B & A \\ 3 \times 4 & 2 \times 3 \\ \hline 3 \times 2 \end{matrix}$

BA = doesn't exist

$AB = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 21 & 8 & 10 & 25 \\ 42 & -14 & 2 & 40 \end{bmatrix}$

$1 \times 1 = 1(5) + 4(5) + (-2)(-2) = 21$
 $1 \times 2 = 1(2) + 4(6) + (-2)(9) = 8$
 $1 \times 3 = 1(8) + 4(4) + (-2)(7) = 10$
 $1 \times 4 = 1(-1) + 4(5) + (-2)(3) = 25$
 $2 \times 1 = 3(5) + 5(3) + (-6)(-2) = 42$
 $2 \times 2 = 3(2) + 5(6) + (-6)(9) = -14$
 $2 \times 3 = 3(8) + 5(4) + (-6)(7) = 2$
 $2 \times 4 = 3(-1) + 5(5) + (-6)(-3) = 40$

Combination

A, B, C, D, E, F

$\frac{6 \times 5 \times 4 \times 3}{1 \times 2 \times 3 \times 4} = \frac{6!}{(6-4)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 360$

of ways to arrange k things in k spots

$n P_k = \frac{n!}{(n-k)!}$

- ① A B C D } order matter
- ② D A B C } two difference form

$\frac{k}{1} \frac{k-1}{2} \frac{k-2}{3} \dots \frac{1}{k} = k!$

$n C_k$ ↓ order doesn't matter 2 form above is same form

$n C_k = \frac{n!}{(n-k)! k!} = \frac{n!}{k!(n-k)!}$

${}^6 C_4 = \binom{6}{4} = \frac{6!}{4!(6-4)!} = \frac{6!}{4!(2!)} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 2 \times 1} = \frac{30}{2} = 15$

${}^{10} C_3 = \binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10!}{3!(7!)} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = \frac{720}{6} = 120$

Practice Questions

Level 1 understanding the concept

- if $n=6, k=2$, is this a valid column? = 1, 0, 1, 0, 0, 0 ✓
- if $n=6, k=2$, is this a valid column? = 1, 1, 1, 0, 0, 0 X
- if $n=6, k=2$, is this a valid column? = 0, 0, 0, 0, 0, 0 X

${}^6 C_2 \binom{6}{2} = \frac{6!}{2!(6-2)!} = \frac{6 \times 5}{2} = 15$

Modulus study

$a \equiv b \pmod{n}$ if and only if $n \mid a-b$ (n divides $a-b$)

a, b and n are integers; $n > 1$

Ex: $5 \equiv 2 \pmod{3}$

$a-b = nk$

$3 \mid 5-2$

$4 \equiv -1 \pmod{5}$

$5 \mid 4 - (-1)$

Ex. Find all possible values of n if $11 \equiv 5 \pmod{n}$

$n \mid 11-5$

$11 \equiv 5 \pmod{6}$



• Modular Arithmetic deals with remainders

• Divide a by n and let the quotient be q and the remainder r , then we have

$a = nq + r$ which implies $a \equiv r \pmod{n}$

Examples

13 divide by 5

$$\begin{array}{r} 2 \\ 5 \overline{)13} \\ \underline{10} \\ 3 \end{array}$$

quotient is 2 (q)

Remainder is 3 (r)

$13 \equiv 3 \pmod{5}$

$2 \equiv -3 \pmod{5}$

$5 \mid 2 - (-3)$

$-8 \equiv 7 \pmod{5}$

$5 \mid -8 - 7$

$5 \mid -15$

$-3 \equiv -8 \pmod{5}$

$5 \mid -8 - (-3)$

$5 \mid -5$

We can use addition, subtraction, and multiplication but division is problematic

Ex: Solve $x^2 + 3 \equiv 0 \pmod{7}$

$7 \equiv 0 \pmod{7}$

$x^2 \equiv -3 \pmod{7}$

$n \equiv 0 \pmod{n}$

$x^2 \equiv 4 \pmod{7}$

$\begin{cases} x \equiv 2 \pmod{7} \\ x \equiv -2 \pmod{7} \end{cases}$

$nk \equiv 0 \pmod{n}$

$\{0, 1, 2, 3, 4, 5, 6\}$

Ex: $10 \equiv 4 \pmod{6}$

$6 \mid 10-4$

$10 \equiv 4 \pmod{6}$

$\downarrow$
 $5 \equiv 2 \pmod{6}$

Ex. $x^2 \equiv 2 \pmod{4} \rightarrow \{0, 1, 2, 3\}$
 $\downarrow \downarrow \downarrow \downarrow$
 $\{0, 1, 0, 1\} \leftarrow x^2$

No solution

What about negative exponents

$3x \equiv 1 \pmod{5}$

$\rightarrow 3x \equiv 6 \pmod{5}$

$3^{-1} \cdot 3x = 3^{-1} \cdot 1$

$x \equiv 2 \pmod{5}$

$x \equiv 3^{-1}$

$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$